

Life Explorer Experiment Guide

Penguin Blubber Experiment

Time

20–25 minutes

Prep Time

5 minutes

Mess Level



Learning Goal

Children discover that a layer of fat (blubber) helps keep animals warm by slowing heat loss in freezing water.

Parent Backpack

Before your child joins you:

- ✓ Fill a bowl halfway with ice and cold water.
- ✓ Make the 'blubber glove' (instructions below).
- ✓ Lay out towels.
- ✓ Queue the penguin video if desired.

Total prep: about 5 minutes.

Materials

- 2 gallon-size zip-top bags
- 1 cup vegetable shortening (Crisco)
- Large bowl
- Ice cubes
- Cold water
- Towels

How to Build the Blubber Glove

1. Turn one zip-top bag inside out.
2. Spread the shortening over the outside of the inside-out bag.
3. Place the second bag over it so the shortening is sealed between the two bags.
4. Your child places their hand inside the inner bag, staying clean while the shortening acts as 'blubber.'

Teacher Script

Say:

'I have a mystery today. Penguins jump into freezing Antarctic water every day. Why don't they freeze? Let's become scientists and investigate.'

Prediction

Ask: 'What do you think will happen when I put my hand in the ice water? What do you think will happen with the blubber glove?' Record the prediction in the Explorer Journal.

Experiment

Round 1: Place a bare hand into the ice water for 10 seconds.

Round 2: Place the hand inside the blubber glove and repeat.

Ask: 'What felt different? Why do you think that happened?'

Teaching Point

Explain briefly: *'Blubber is a thick layer of fat that helps keep body heat from escaping. Penguins, seals, and whales all have ways to stay warm in extremely cold environments.'*

Vocabulary

blubber • insulation • temperature • survive • freezing

Journal Prompt

Draw the experiment and finish the sentence: *'I discovered...'*

Keep Exploring

If your child asks about seals, whales, or polar bears, follow that curiosity—it may become tomorrow's lesson.

Helpful Resources

PBS LearningMedia – Animal Insulation Activity (teacher guide)

https://static.pbslearningmedia.org/media/media_files/3d4ce992-c70e-4a6b-9178-20bcb28ca4d7/e8c63031-db26-4c6f-a8e4-adc4f51c9e95.pdf

PBS Nature – Penguins: Meet the Family

<https://www.pbs.org/video/penguins-meet-the-family-bzdmpa/>

Teacher Tip: Open these resources before the lesson so there is no waiting while your child watches.